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Modeling and evaluation of the influence of local bicycle infrastructure on the usage behavior of bicycles and pedelecs

Bachelor thesis

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1 Introduction

Climate change is one of the greatest challenges of our time, with the transport sector being one of the main contributors to greenhouse gas emissions (Boucher, 2019, p. 17). In this context, bicycles are a low-emission alternative to motorized means of transport and can thus reduce CO₂ emissions in the transport sector (Sallis et al., 2016, p. 2216). This could make a significant contribution to climate protection (Pucher & Buehler, 2008, p. 496).

Furthermore, cycling promotes efficiency and quality of life in cities. This not only leads to a lower environmental impact, but also enables a more efficient use of urban space. It also reduces congestion and dependence on cars, which means that bicycle-friendly cities support sustainable urban development (Litman, 2013, p. 72). Furthermore, the promotion of physical activity is a key public health concern. Regular cycling improves fitness and reduces body weight. It also reduces the risk of numerous chronic and mental illnesses (Brown et al., 2013, p. 233).

However, the use of bicycles is primarily limited to shorter distances. Due to the physical effort and time required to ride a bicycle compared to motorized vehicles, longer trips are often less feasible (Gallo & Marinelli, 2020, p. 7-8). Pedelects (pedal electric cycles) expand the potential of cycling by providing electric assistance when pedaling. This support enables users to cover longer distances and overcome topographical challenges more easily (Fishman & Cherry, 2016, p. 9).

In view of the fact that an adequate bicycle infrastructure is a fundamental prerequisite for the use of bicycles and pedelects, this research focuses on the investigation of the local bicycle infrastructure and its effects on usage behavior. This is achieved with a literature analysis of the current state of research as well as the evaluation of geodata and survey results.

2 State of the art Science

Whether people ultimately choose bicycles and pedelecs as a means of transportation can have several factors. For example, social and psychological factors such as the social environment or personal habits can influence attitudes towards cycling (Willis et al., 2015, p. 565). Workplace culture also plays a role in commuting, especially when it comes to the choice of transportation. The fact that many colleagues¹ use bicycles to get to work increases the likelihood that others will also choose to cycle (Heinen et al., 2013, p. 23). Among other things, the distance to work is decisive. The probability of cycling decreases by around 8 % with each additional mile (Willis et al., 2015, p. 577). Furthermore, seasonal weather conditions such as snow, rain and wind have an impact on the likelihood of using the bicycle as a means of transportation (Sears et al., 2012, p. 109).

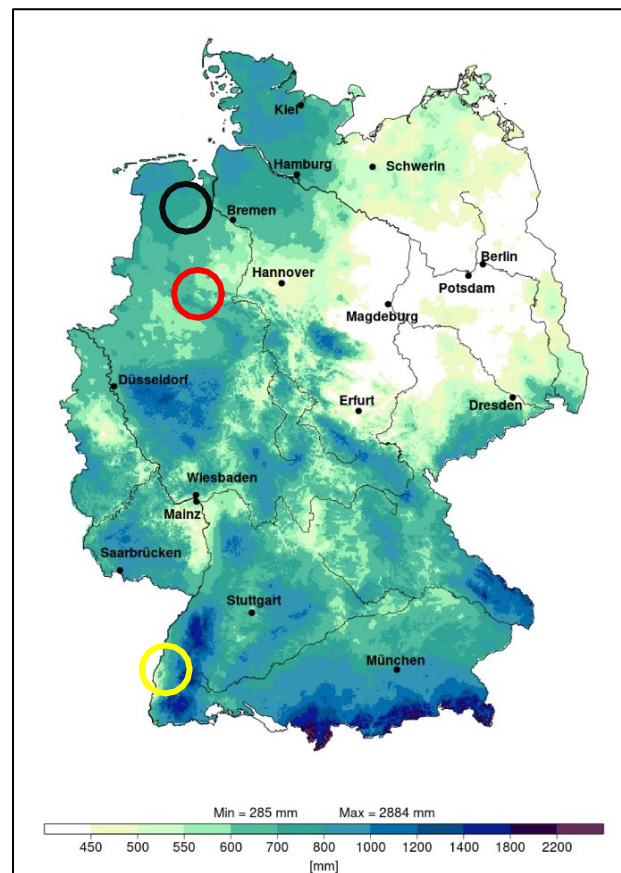


Figure 1: Annual precipitation in Germany in 2022 [in mm] according to Deutscher Wetterdienst (2023).

¹For better readability, the generic masculine is used in the text. However, all genders are always meant.

Looking at the annual amount of precipitation in Germany for the year 2022 in Figure 1 shows that regions with higher precipitation values such as Oldenburg (black circle), Münster (red circle) and Freiburg (yellow circle) continue to record high levels of bicycle use (Bocksch, 2020, p. 1). This observation suggests that the weather does not have the decisive influence on the choice of cycling as a means of transportation (Nankervis, 1999, p. 430). Rather, the studies indicate that other factors have a stronger influence on the use of bicycles. A large number of studies show that the availability and quality of local bicycle infrastructure in particular has a significant influence on bicycle use (Pucher & Buehler, 2008, p. 495; Teixeira et al., 2020, p. 1; Wysling & Purves, 2022, p. 2).

Cycling infrastructure refers to all the facilities and features of an area that promote cycling as a safe, attractive and practical transport option. To evaluate cycling infrastructure, scientific studies identify various key indicators (see Table 14 in Appendix A.1). These criteria were mainly analyzed in urban areas such as Graz (Krenn et al., 2015, p. 451), Barcelona (Codina et al., 2022, p. 1) and Vancouver (Winters et al., 2013, p. 865). Methodologically, the studies are based either on local surveys (Codina et al., 2022, p. 1; Gullón et al., 2015, p. 925), the direct investigation of cycle routes (Krenn et al., 2015, p. 451) or on comprehensive literature analyses to determine the indicators (Hartanto et al., 2017, p. 1; Ito & Biljecki, 2021, p. 3; Wysling & Purves, 2022, p. 2). Expert surveys are used to determine the weighting of indicators (Arellana et al., 2020, p. 317; Hardinghaus et al., 2021, p. 8; Stinson & Bhat, 2003, p. 107; Winters et al., 2013, p. 865), literature analyses (Wysling & Purves, 2022, p. 2) and a simplified weighting method in which all indicators are assigned the same weight (Hartanto et al., 2017, p. 23; Ito & Biljecki, 2021, p. 9).

The objectives of the studies vary, including the analysis of the transit infrastructure in Arnhem (Hartanto et al., 2017, p. 1), the creation of local bikeability indices² to visualize and evaluate the cycling infrastructure (Gullón et al., 2015, p. 923; Ito & Biljecki, 2021, p. 15; Krenn et al., 2015, p. 451; Winters et al., 2013, p. 874), the localization of expansion opportunities (Arellana et al.,

²Evaluation system for the cycling infrastructure of an area.

2020, p. 310; Wysling & Purves, 2022, p. 1), the investigation of commuters' route choice (Stinson & Bhat, 2003, p. 107) and the investigation of the correlation between cycling infrastructure and cycling behavior at the urban level (Codina et al., 2022, p. 7; Hardinghaus et al., 2021, p. 6).

Various methods are used for data collection. Geographic information systems (GIS) are used to collect road and terrain data and to visualize the Bikeability Index (Arellana et al., 2020, p. 312; Codina et al., 2022, p. 4; Hartanto et al., 2017, p. 5; Krenn et al., 2015, p. 451; Winters et al., 2013, p. 873). Google Street View is also used to enable the assessment of cycling infrastructure based on images (Gullón et al., 2015, p. 923; Ito & Biljecki, 2021, p. 1). A supplementary data collection method is based on the road data from OpenStreetMap (OSM) (Hardinghaus et al., 2021, p. 1; Ito & Biljecki, 2021, p. 1; Wysling & Purves, 2022, S. 1).

Hardinghaus et al. (2021, p. 1) developed a Bikeability Index using road data from OSM, which is used specifically to evaluate Berlin's cycling infrastructure. In future research, they recommend taking into account additional factors such as topography and surface conditions (Hardinghaus et al., 2021, pp. 12-13). Other limitations include the focus of the Bikeability Indices on specific areas and the challenges posed by Google Street View data sources that are sometimes outdated or not fully representative (Gullón et al., 2015, p. 936; Ito & Biljecki, 2021, p. 17). The non-consideration of green spaces (Gullón et al., 2015, p. 936), connectivity, underground quality (Wysling & Purves, 2022, p. 9) and a simplified weighting of the indicators (Hartanto et al., 2017, p. 23; Ito & Biljecki, 2021, p. 9) can also impair the accuracy of the assessment.

In view of the fact that the focus of research on bicycle infrastructure has so far been primarily on urban areas and classic bicycles, this bachelor thesis aims to close the existing research gap regarding the influence of bicycle infrastructure on the usage behavior of bicycles and pedelecs in Germany in urban and rural areas. This is to be achieved through a detailed modeling and evaluation of the correlation between bicycle infrastructure and usage behavior.

take place. Against this background, the central research question is formulated as follows:

To what extent does bicycle infrastructure influence the use of bicycles and pedelecs in urban and rural regions of Germany?

In order to answer the defined research question, systematic work steps are carried out in four main phases. In the first phase, the indicators for urban and rural areas are selected and then weighted through a literature analysis in order to develop a Bikeability Index. In the second phase, the Bikeability Index is implemented on the basis of OSM data. This also includes the geographical distribution into grid cells for spatial analyses. The Bikeability Index will then be compared with existing approaches from Europe and America. In the third phase, the Bikeability Index is applied to the Federal Republic of Germany as part of a case study and evaluated using a logistic regression analysis. In the final phase, the results and limitations are discussed.

3 Methodology and derivation of the Bikeability Index

The development process of a Bikeability Index starts with a literature analysis (see Figure 2). The aim is to identify relevant indicators for urban and rural areas that can be used to assess cycling infrastructure. These indicators are selected on the basis of their importance and their weighting is determined using the literature. In addition, existing bikeability indices are presented, which are used in the next chapter to validate the newly developed Bikeability Index.

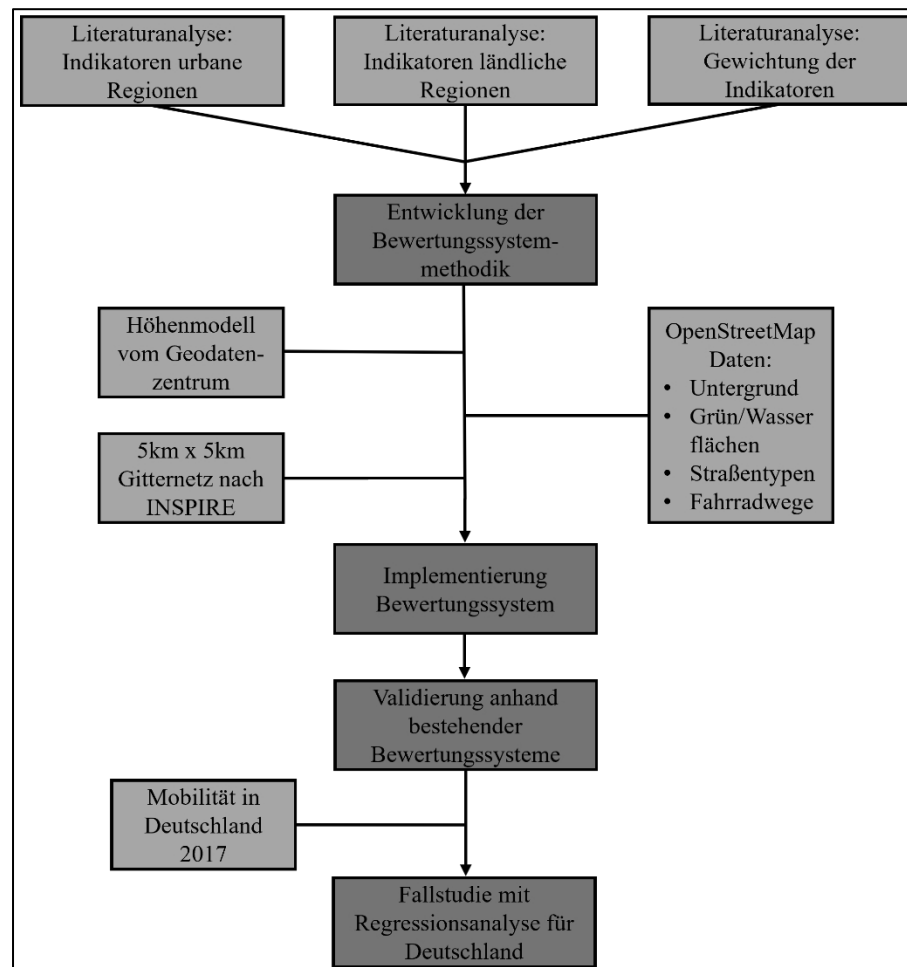


Figure 2: Visualization of the methodology: analysis steps (dark grey) and data sources (light grey).

3.1 Selection Indicators

When evaluating cycling infrastructure, it is crucial to identify the relevant influencing factors. Based on the current state of research, the factors can be summarized in six categories: Proportion of traffic-calmed roads, proportion of cycle paths along

main roads, connectivity of cycle paths, topography, surface quality and proportion of green and water areas (Arellana et al, 2020; Gullón et al, 2015; Hardinghaus et al, 2021; Hartanto et al, 2017; Ito & Biljecki, 2021; Krenn et al, 2015; Steinacker et al, 2022; Valenzuela et al, 2022; Winters et al, 2013; Wysling & Purves, 2022). These are described and derived below.

Proportion of traffic-calmed and bicycle-friendly streets:

Huber et al. (2021, p. 1) show that speed limits have a positive influence on the likelihood of cyclists choosing a route. Higher speed limits are often associated with main traffic routes in the city (Wysling & Purves, 2022). In addition, roads with less traffic and a lower speed limit are perceived as safer and therefore preferred by cyclists (Chan & Cooper, 2019, p. 4). Steinacker et al. (2022, p. 661) shows that traffic congestion increases from residential streets to main roads. Residential streets and streets with lower speed limits can therefore be considered traffic-calmed.

Krenn et al. (2015, p. 453) and Winters et al. (2013, p. 872) assess the cycle-friendly roads based on the total length of all cycle paths, including separated cycle paths and cycle lanes along the roads. Comparability between rural and urban areas is promoted by calculating the proportion of traffic-calmed and bicycle-friendly roads to the total road network (Hardinghaus et al., 2021, p. 5).

Ferster et al. (2020, p. 1) and Huber et al. (2021, p. 4) show that separate cycle paths allow cyclists to avoid motorized traffic, which in turn significantly increases the safety and attractiveness of cycling. The study by Codina et al. (2022, p. 4) provides additional weighting by assigning greater importance to separate cycle paths. The evaluation methodology is adopted, in which separate cycle paths are taken into account with a double weighting (Walk Score, 2024). Due to the comparability between rural and urban areas, the subdivision according to Hardinghaus et al. (2021, p. 10) is adopted. The scale ranges from 0 points for a share of 0 % to 10 points for a share of over 0.87 % (see Table 1).

Proportion of traffic-calming and bicycle-friendly roads [%]	0 %	0 %	26 %	58 %	75 %	87 %
		-	-	-	-	-
		26 %	58 %	75 %	87 %	100 %
Points	0	2	4	6	8	10

Table 1: Rating scale: Proportion of traffic-calmed streets according to Hardinghaus et al. (2021).

Proportion of cycle paths on main roads:

Marshall and Garrick (2011, p. 26) show that comparable cities with lower cyclist fatality rates often have fewer main roads. In addition, Valenzuela et al. (2022, p. 4) and Wysling & Purves (2022, p. 4), together with the studies from the previous chapter, show that cyclists prefer clearly separated routes from motorized traffic. Cyclists tend to avoid main roads without lane separation, which is why cycle paths next to main roads are particularly important (Krenn et al., p. 455; Pucher & Buehler, p. 523; Steinacker et al., 2022, p. 659).

Krenn et al. (2015, p. 453) assess the length of all main roads without parallel cycle lanes as part of their analysis of cycling infrastructure. In comparison, Hardinghaus et al. (2021, p. 6) set the length of main roads with separate cycle lanes and cycle tracks in relation to the total length of all main roads. This approach enables a comparable assessment of cycle lanes on main roads across different areas. This is done by not using the total length as the decisive factor, which would put areas with a larger road network at a disadvantage. Hardinghaus et al. (2021, p. 10) rate cycle path coverage along main roads on a scale from 0 % (no coverage) to 100 % (full coverage). Table 2 shows the point scale from 0 to 10 points.

Proportion of cycle paths on main roads [%]	0 %	0 %	21 %	47 %	67 %	86 %
		-	-	-	-	-
		21 %	47 %	67 %	86 %	100 %
Points	0	2	4	6	8	10

Table 2: Rating scale: Proportion of cycle paths on main roads according to Hardinghaus et al. (2021).

Connectivity of cycle paths:

Valenzuela et al. (2022, p. 10) and Codina et al. (2022, p. 4) state that the connectivity of the cycle paths is central, as has a high connectivity

offers cyclists more route options. This increases the possibility of choosing routes according to individual preferences. Furthermore, Winters et al. (2013, p. 869) and Ito and Biljecki (2021, p. 3) show that the connectivity, accessibility and directness of the routes of the bicycle network reflect This factor shows how effectively cyclists can get from one point to another without having to make detours on motorized roads.

Connectivity as an indicator of cycling infrastructure is uniformly measured by the number of intersections within a given study area. An approach that Codina et al. (2022, p. 4), Winters et al. (2013, p. 872), Ito and Biljecki (2021, p. 3) and Hardinghaus et al. (2021, p. 5) consistently apply in their studies.

The size of the study areas varies, for example Hardinghaus et al. (2021, p. 5) measures the number of cycle path crossings per km² and Winters et al. (2013, p. 872) in a 400 meter radius. These different approaches can be standardized using a scaling factor and adapted to different study areas. In the present research work, the scale of Winters et al. is adopted, as it has a finer scaling of the crossings (see Table 3). This scale ranges from 0 points (no crossings) to 10 points (over 2723 crossings).

Number of crossings	0	1 – 49	50 – 149	150 – 298	299 – 497	498 – 746	747 – 994	995 – 1243	1244 – 1492	1493 – 2723	> 2723
Points	0	1	2	3	4	5	6	7	8	9	10

Table 3: Rating scale: Connectivity according to Winters et al. (2013).

Topography:

Research by Huber et al. (2021, p. 4) and Codina et al. (2022, p. 4) proves that the incline has a significant influence on driving behavior. In contrast to Arellana et al. (2020, p. 324), who found that the gradient has a minor influence. To precisely record and analyze the topography, Winters et al. (2013, p. 873), Codina et al. (2022, p. 4), Krenn et al. (2015, p. 453), Ito and Biljecki (2021, p. 7) and Wysling and Purves (2022, p. 3) digital elevation models. The range of methods for height determination extends from a radius of 100 meters (Krenn et al., 2015, p. 453) to a radius of 2400 meters (Hartanto et al., 2017, p. 8).

Hardinghaus et al. (2021, p. 13) show that future research should integrate topography as an indicator to obtain a more comprehensive picture of cycling infrastructure. This underlines the need to include topographical data in the assessment, as further emphasized by Krenn et al. (2015, p.453). Gradient classes are defined in order to incorporate topography into analyses. The use of a clear and quantifiable scale to assess the gradient, such as the subdivision according to Winters et al. (2013, p. 873), enables a differentiated assessment of the topography. For this reason, this scale is integrated into the Bikeability Index. In this context, it can be seen that there is an inverse relationship between gradient and bicycle use (see Table 4).

Gradient [%]	> 20 %	< 10 %	< 5 %	< 2 %	< 0,5 %	0 %
Points	0	2	4	6	8	10

Table 4: Rating scale: Topography according to Winters et al. (2013).

Surface finish:

The importance of surface quality as a decisive factor for cycling route choice is emphasized by Stinson and Bhat (2003, p. 108) and Huber et al. (2021, p. 1). Both identify the surface condition as a factor influencing the route decision. Hartanto et al. (2017, p. 13) and Hardinghaus and Nieland (2021, p. 4) expand on these findings by evaluating surface quality as a key safety factor and highlighting specific preferences such as avoiding cobblestones. The importance of surface quality is also emphasized by Arellana et al. (2020, p. 315), who highlight the surface type as a central aspect for cycling comfort and classify it as an integral part of the cycling infrastructure. On the other hand, Mertens et al. (2016, p. 11) show that surface quality is of limited importance compared to other factors.

The methodical evaluation of the surface condition is supplemented by Ito and Biljecki (2021, p. 11) and Hardinghaus and Nieland (2021, p. 4), where Ito and Biljecki present a specific classification of the surface condition. This classification is supplemented by Hardinghaus and Nieland with a detailed rating scale. It ranges from 0 points for less

preferred surfaces such as cobblestones up to 10 points for high-quality surfaces such as asphalt and concrete (see Table 5). In the present research work, this evaluation of the surface condition is adopted.

Surface finish	Cobblestone pavement, sand	Earth, wood, crushed stone, gravel, grass	Plaster	Concrete	Asphalt
Points	0	4	6	8	10

Table 5: Rating scale: Surface texture according to Hardinghaus et al. (2021), Ito and Biljecki (2021).

Proportion of green and water areas:

The studies by Gullón et al. (2015, p. 925) and Hartanto et al. (2017, p. 10) emphasize the importance of the environment, especially green and water areas, for the quality of cycling infrastructure. These studies consider the natural environment to be a key factor that increases the comfort and attractiveness of cycling. A distinction can be made between rural and urban areas. Rural areas offer a natural landscape that makes cycling more attractive, while urban areas have to cope with the integration of green and water areas within the built environment. Ito and Biljecki (2021, p. 11) take a similar perspective but extend the analysis by examining the balance between green spaces, buildings and water to paint a comprehensive picture of the natural and built environment. While Krenn et al. (2015, p. 451) focus on the total area of public green spaces without directly considering the paths, Hardinghaus et al. (2021, p. 6) calculate the percentage of green and water areas in relation to the total area of the areas studied. This enables a differentiated assessment of the green and water areas.

Hardinghaus et al. (2021, p. 10) use a rating scale that takes into account the percentage of green and water areas in a 1 x 1 km grid cell. The scale according to Hardinghaus et al. is adapted, as the original scale awards the maximum score of 10 points from 61%. An additional evaluation level is introduced for this purpose. This should help to take greater account of the special features of rural regions. The revised scale now ranges from 0 points for a share of less than 9 % of green spaces to 10 points for a share of less than 10 %.

and water areas up to 10 points for a share of over 81 % (see Table 6).

Share of green and water areas [%]	< 9 %	9 % - 22 %	22 % - 41 %	41 % - 61 %	61 % - 81 %	81 % - 100 %
Points	0	2	4	6	8	10

Table 6: Rating scale: Proportion of green and water areas according to Hardinghaus et al. (2021).

Other indicators:

In addition to the indicators already mentioned, there are a number of other factors that can influence the use of bicycles. However, these are not used in our model due to their subordinate relevance. On the one hand, the availability of bicycle parking spaces (Codina et al., 2022, p. 4; Stinson & Bhat, 2003, p. 108) and bicycle repair shops play a role (Codina et al., 2022, p. 4; Hardinghaus et al., 2021, p. 3). On the other hand, environmental factors such as air pollution and traffic noise also play a role. This is partly taken into account in our model, as traffic-calmed streets, for example, have lower air pollution and less traffic noise (Valenzuela et al., 2022, p. 8). The quality of connections to other modes of transport can also have an influence, as good transit availability makes it easy to switch from one mode of transport to another (Hartanto et al., 2017, p. 1). Safety for cyclists is an important factor and is taken into account in this study by considering separated cycle paths along the main roads. However, details such as the width and color markings of the cycle paths can play an additional role (Steinacker et al., 2022, p. 655).

3.2 Weighting of the indicators

In the chapter on the current state of research, it was shown that various approaches have emerged in research on the weighting of indicators. In this research paper, the weighting is derived from the literature. Hardinghaus et al. (2021, p. 4) highlight the importance of cycle paths along main roads and the connectivity of cycle paths as particularly relevant areas. Gullón et al. (2015, p. 925) identifies the following as the most important

positive factor is the presence of a cycle path next to and on the road. In addition, Stinson and Bhat (2003, p. 111) and Codina et al. (2022, p. 4) point out that the biggest negative factor for the route decision is the presence of main roads and the associated motorized traffic. Based on various studies, an average value is calculated for each indicator and then adjusted so that the overall weighting is 100%. This weighting integrates the perspectives of the individual studies into a Bikeability Index (see Table 7). In the development process, the indicators from the studies are transferred to the selected indicators of the Bikeability Index and the weightings are adjusted accordingly.

Index	Cycle paths along main roads	Traffic-calmed streets	Connectivity	Green and water areas	Topography	Underground
Stinson and Bhat (2003)	30 %	30 %	10 %	x	10 %	20 %
Gullón et al. (2015)	30 %	30 %	x	10 %	20 %	10 %
Krenn et al. (2015)	40 %	20 %	x	20 %	20 %	x
Hardinghaus et al. (2021)	28 %	17 %	23 %	16 %	x	x
Arellana et al. (2020)	25 %	20 %	10 %	x	10 %	15 %
Codina et al. (2022)	27,5 %	27,5 %	15 %	x	20 %	x
Interval	25 %	17 %	10 %	10 %	10 %	10 %
	40 %	30 %	23 %	20 %	20 %	20 %
Average	~ 26 %	~ 21 %	~ 13 %	~ 13 %	~ 14 %	~ 13 %

Table 7: Weighting of the Bikeability Index.

3.3 Comparative indices

Due to their simple structure and reproducible results, the indices by Krenn et al. (2015) and Winters et al. (2013) are considered reliably established studies for the evaluation of bicycle infrastructure in urban areas and represent the basis for future research. Furthermore, they were selected due to their study areas in Europe and America, which

enables a continental comparison (Krenn et al., 2015, p. 451; Winters et al., 2013, p. 868). They serve as a basis for comparison in order to critically reflect on the Bikeability Index created for the research work and to check its validity.

The study by Winters et al. (2013, p. 865) aims to develop a comprehensive bikeability index for Vancouver by examining mobility behavior and conducting an expert survey. The study area covers a grid cell with a size of 400 meters. Key indicators include the density and quality of bicycle routes, the separation of bicycle lanes from motorized traffic, the connectivity of bicycle-friendly streets and the topography (Winters et al., 2013, p. 872). Another indicator is the density of destinations, but this is not taken into account in this analysis as specific destinations are not defined.

The research by Krenn et al. (2015, p. 451) is dedicated to the detailed analysis of cycling trips in Graz. The aim of these analyses is to determine specific indicators of cycling infrastructure that influence cyclists' route choices. Their investigation includes various elements of cycling infrastructure, including separated cycle paths, main roads without accompanying cycling infrastructure, green and water areas, topography and land use. Land use was not identified as an important factor, which is why this indicator is not taken into account (Krenn et al., 2015, p. 454). The selection of a grid cell size of at least 250 meters is based on previous research, which showed that cyclists on average accept a distance of 250 meters (Krenn et al., 2015, p. 453).

In order to ensure the comparability of these indices with the Bikeability Index, scaling to the same grid cell size is necessary. Equation (1) is used to calculate the scaling factor. Initially

the area of the grid cell under consideration [G] is determined and then divided by the divided by the grid cell area [G_S] specified for the study. The scaling factor is applied to the number of crossings, for example, in order to ensure a uniform evaluation.

$$\text{Skalierungsfaktor} = \frac{G}{G_S} \quad (1)$$

4 Application and validation of the Bikeability Index

The Bikeability Index is applied using OSM data. The Python library OSMnx is used to calculate the indicators. The following section explains the implementation of the indicators, including the specific functions and equations. Finally, the Bikeability Index is validated with the existing indices using various areas.

4.1 Description of the OpenStreetMap data

OpenStreetMap is a data source for bicycle infrastructure research. As a comprehensive, globally available data collection, OSM covers transportation infrastructure as well as urban and rural development. The maps are particularly detailed and contain a wealth of information about paths, roads, cafés, train stations and more (OpenStreetMap, 2022). Another advantage of OSM is the high level of up-to-dateness, which is made possible by the continuous participation of over 11 million people worldwide. This participation ensures that changes in the real world can be quickly updated in the database (Ferster et al., 2020, p. 8). A comparison shows that the total length of cycle paths in urban areas corresponds to the official data provided by municipal authorities (Ferster et al., 2020, p. 9). OSM data is available under the Open Database License (ODbL), which allows the use of this data for a variety of applications free of charge (OpenStreetMap: ODbL, 2022).

In order to simplify the collection, creation and analysis of road networks using OSM data, the Python library OSMnx is used in this research work. When queried, OSMnx typically returns a GeoDataFrame consisting of nodes and edges. The nodes represent a specific point on the earth and represent, for example, a traffic light. The edges contain details about the paths such as their geometry (coordinates, gradient and length), type (e.g. cycle path or road) and other OSM-specific tags that provide additional information such as the name, surface and use of the paths (Boeing, 2023, p. 14). A detailed description of the data model and the different types of

Road networks can be found in Appendix A.2. There are no official standards on how to tag, but OSM has published a recommendation (OpenStreetMap Wiki: Map Features, 2022). The challenge is to define and apply the tags in such a way that they accurately reflect reality. The tags, which are essential for the evaluation of cycling infrastructure, are described in the next section.

Class	Description	OSM tags
Primary roads	Main road, but with lower capacity and speed.	highway = primary highway= primary_link
Secondary roads	Small main road that mediates traffic between the primary roads.	highway = secondary highway= secondary_link
Tertiary roads	Side roads that Route traffic from the local roads to the main traffic arteries.	highway = tertiary highway= tertiary_link
Municipal roads	Publicly accessible side road. Typically without a center line.	highway= unclassified
Local roads	Roads with access to residential properties and other end points.	highway= living_street highway = residential
Service road	Field and forest road mainly used for agriculture and forestry.	highway= track
Traffic-free routes	Paths that are excluded from motorized traffic.	highway= cycleway highway = path + * highway= footway/pedestrian+ *
Service roads	Routes with vehicle access to public/private facilities.	highway= service

Table 8: Categorization of streets from tags in OSM according to Chan and Cooper (2019), OpenStreetMap Wiki: Map Features (2022).

Relevant values for the *"highway"* tag indicate road types ranging from multi-lane *"primary"* roads to subordinate *"service"* roads. (Chan & Cooper, 2019, p. 10). For the classification, Table 8 provides an overview of the relevant *"highway"* tags in this work. Steinacker et al. (2022, p. 661) shows that the expected traffic load is steadily increasing from *residential* to *primary* roads. This is also an indication of the increasing traffic speed (Chan & Cooper, 2019, p. 4).

The tag *"highway=service"* refers to roads that are intended for vehicle access to facilities such as gas stations and parking lots. Such roads are usually not part of the public road network and are sometimes not accessible to the general public (OpenStreetMap Wiki: Map Features,

2022). These routes are therefore removed from the geodata frame and are not included in the evaluation.

With regard to cycle paths, OSM concentrates on areas that are either designated as general roads or specifically as cycle paths. These cycle paths can either be marked as a separate path ("*highway= cycleway*" or "*highway= path+ bicycle= yes/designated*") or alternatively added as a marking on an existing road ("*highway= *+ cycleway = lane*") (Ferster et al., 2020, p. 3). Table 9 shows other possible combinations for the marking of cycle paths. A cycle path that has its own marking but runs alongside a road and is part of the road network is marked as "*cycleway=separate*" (OpenStreetMap Wiki: Map Features, 2022).

Kind	OSM tags (*different road classes)
Insulated cycle path	highway= cycleway
Separate cycle path next to a road	highway= *+ cycleway= track/opposite_track
Cycle path on the roadway	highway= *+ cycleway= lane
Path (bicycle only or multi-purpose)	highway= path+ bicycle= yes/designated highway = footway + bicycle = yes highway= *+ bicycle= use_sidepath/designated highway = cycleway
Bicycle road	bicycle_road= yes

Table 9: Cycle path types and their OSM tags according to Ferster et al. (2020).

4.2 Implementation of the indicators

To prepare the analysis for the case study in the later chapter, grid cells with a size of 5 x 5 km are selected. This makes it possible to assign a specific bicycle infrastructure assessment to each grid cell. Depending on the needs, the size of the grid cells can be adjusted to allow for a more detailed or broader analysis. For each 5 x 5 km grid cell, the cycling infrastructure is assessed based on the defined indicators and their weightings. How these indicators are implemented is explained below.

Proportion of traffic-calmed streets (V):

In order to include all traffic-calmed roads (S), the tertiary roads, municipal roads, local roads, farm tracks and traffic-free roads are taken into account. paths from Table 8 are included. In addition, cycle lanes and exclusive cycle paths are included in the analysis, with the length of the isolated cycle paths being weighted twice. Finally, equation (2) is used to calculate the proportion of traffic-calmed roads (V).

$$V = \frac{\sum \text{Länge } (S)}{\sum \text{Länge (alle Straßensegmente)}} \times 100. \quad (2)$$

The result of this calculation is used to assign the corresponding number of points from Table 1.

Proportion of cycle paths on main roads (R):

The tags for primary and secondary roads are used to define main roads. To identify the cycle paths on the main roads, the cycle path types listed in Table 9 are used in conjunction with the *"highway"* tag. The ratio of the lengths of these cycle paths along the main roads (F) to the total length of all main roads (H) is calculated using equation (3). The corresponding score is then determined from Table 2.

$$R = \frac{\sum \text{Länge } (F)}{\sum \text{Länge } (H)} \times 100. \quad (3)$$

If there is no main road in a grid cell, this grid cell is skipped and not evaluated, as otherwise there may be distortions in the Bikeability Index.

Connectivity (K):

To measure connectivity, all tags indicating separate paths are first removed so that only cycle paths are taken into account and not the associated road. To measure this, the different types of cycle paths are taken from Table 9. The number of intersections involving at least two roads is then recorded using an OSMnx function (Boeing, 2017, p. 134). The corresponding score is then calculated from the number of intersections according to Table 3.

Topography (T):

First, a contour grid file of Germany and the OSMnx function are used to determine the gradient in percent for each road segment (Boeing, 2017, p. 134). In the GeoDataFrame, road segments are displayed for each direction with two-way trafficability. To prevent hilly areas from distorting the calculation of the total gradient due to negative gradient values, road segments with negative gradients are excluded from the gradient calculation. According to Table 4, each road segment receives a rating based on the gradient. The gradient of each segment is weighted, taking into account its length, to finally determine a weighted score (T) for each grid cell. Equation (4) is used here.

$$T = \frac{\sum \text{Punktzahl}(\text{Segment}) \times \text{Länge}(\text{Segment})}{\sum \text{Länge}(\text{Segment})}. \quad (4)$$

Surface finish (O):

In OSM, "surface" tags describe the material of paths such as roads, footpaths and cycle paths. These tags range from general classifications such as "paved" or "unpaved" to more specific types such as "asphalt", "cobblestone", "ground" (earth) and "concrete" (concrete). Surface tags provide information about the usability of paths for bicycles and other means of transportation (OpenStreetMap Wiki: Map Features, 2022).

According to Table 5, each road segment is given a score based on the condition of the road surface. The weighted score is then

(O) is determined for each grid cell using equation (5). If the value for the un
If there is no background for a segment, this is not included in the calculation.

$$O = \frac{\sum \text{Punktzahl}(\text{Segment}) \times \text{Länge}(\text{Segment})}{\sum \text{Länge}(\text{Segment})}. \quad (5)$$

Proportion of green and water areas (G):

In OSM, tags such as "leisure=park", "landuse=grass", "landuse=forest" and "natural=aquatic" are used to identify green spaces, forests, meadows and water landscapes. These tags are essential for the mapping of

near-natural areas (OpenStreetMap Wiki: Map Features, 2022). This can result in overlapping areas with different labels. For example, small forest areas within parks can consist of two overlapping polygons (Hardinghaus et al., 2021, p. 6). A function of OSMnx is used to identify green and water areas. Equation (6) is then applied to calculate the ratio of these areas to the total area of the grid cell.

$$G = \frac{\text{Grün- und Wasserflächen} [km^2]}{\text{Gesamtfläche} [km^2]} \times 100. \quad (6)$$

Finally, the final evaluation of the Bikeability Index (E) of a grid cell is calculated by combining the indicators according to the weighting determined in Table 7. Equation (7) is used for this.

$$E = 0.26V + 0.21R + 0.13K + 0.14T + 0.13O + 0.13G. \quad (7)$$

4.3 Validation with Comparative indices

To enable comparability, the study areas are first adjusted with a scaling factor according to equation (1). To validate the applicability and informative value of the developed Bikeability Index, different regions are specifically selected for the comparison of the indices. Metropolitan regions such as Berlin and Hamburg are used for the analysis in densely populated urban areas with complex transportation requirements. In contrast, cycling cities such as Freiburg, Oldenburg and Münster were selected for their exemplary cycling infrastructure in order to test the effectiveness of the index in cycling-friendly environments. Industrial cities such as Essen and Stuttgart enable an investigation in contexts that are traditionally characterized by a dominant car infrastructure. Finally, villages and rural areas provide the basis for evaluating the relevance and applicability of the developed index in areas with a lower population density.

Table 10 shows that cycling cities receive the highest score in the developed index. This differs from the existing indices (Krenn and Winters), in which large cities also achieve high scores. The similarity of the ratings for large cities between the developed index and the Krenn Index

indicates a comparable assessment of this environment. Industrial cities are clearly downgraded in all three indices. The developed index shows a wider assessment range in rural areas. This could confirm the ability to capture cycling infrastructure in a more differentiated way and thus enable applicability beyond urban areas.

	Krenn Index	Winters Index	Developed index
Fribourg	8,30	8,76	8,44
Minster	7,98	9,65	8,37
Oldenburg	7,42	9,50	8,15
Berlin	7,67	9,04	7,66
Hamburg	7,50	9,37	7,65
Food	6,62	6,74	5,51
Stuttgart	6,27	6,85	5,64
Village in the Allgäu	5,56	2,85	4,73
Village near Freiburg	5,38	3,34	6,30
Rural area (Brandenburg)	5,99	2,79	5,10
Rural area (Bavaria)	5,90	2,84	7,62

Table 10: Evaluation of the various indices based on different regions.

5 Case study: Influence of the bikeability score on mobility in Germany

The case study on the influence of the bikeability score on mobility in Germany is based on the "Mobility in Germany 2017" (MiD) data set. The study serves to investigate the usage behavior of bicycles and pedelecs in Germany. First, the mobility data is described, followed by a descriptive analysis of the data set. The usage behavior of bicycles and pedelecs is then illustrated. A logistic regression analysis is used to identify and discuss the correlations between bicycle and pedelec use and the bicycle infrastructure in urban and rural areas.

5.1 Description of the mobility data

The MiD study was commissioned by the Federal Ministry of Transport and Digital Infrastructure and conducted by the infas Institute for Applied Social Sciences in cooperation with the German Aerospace Center (Institute of Transport Research), IVT Research GmbH and infas 360 GmbH (Eggs et al., 2019, p. 13). The field phase of the survey ran from May 2016 to September 2017 (Eggs et al., 2019, p. 15). The total sample comprised 156,420 households, with 33,389 households in the nationwide basic sample and 123,031 in regional in-depth samples⁽³⁾. A total of 316,361 people were surveyed, who provided information on a total of 960,919 journeys made.

On the one hand, personal information of the participants such as gender, age, economic situation, level of education, household type, available means of transport and their usage habits are used to analyze bicycle usage behavior. On the other hand, data is collected on the journeys made, including the means of transport used on the reference date, weather conditions, time of year, purpose of the journey, distance and number of passengers. In addition, 5 x 5 km address data is used, as the detailed resolution of the start and end coordinates is only available for a small proportion of the trips.

³Regional in-depth samples refer to additional, detailed surveys in specific regions.

routes are recorded. This limits the possibility of detailed route tracking.

5.2 Initial data validation and visualization

In order to analyze potential distortions in the MiD data set, an investigation is carried out with regard to various demographic characteristics. These include the average age, gender distribution and ownership of car driving licenses. These characteristics are compared with external data provided by Statista and the TÜV association. Figure 3 shows an overview of the demographic characteristics.

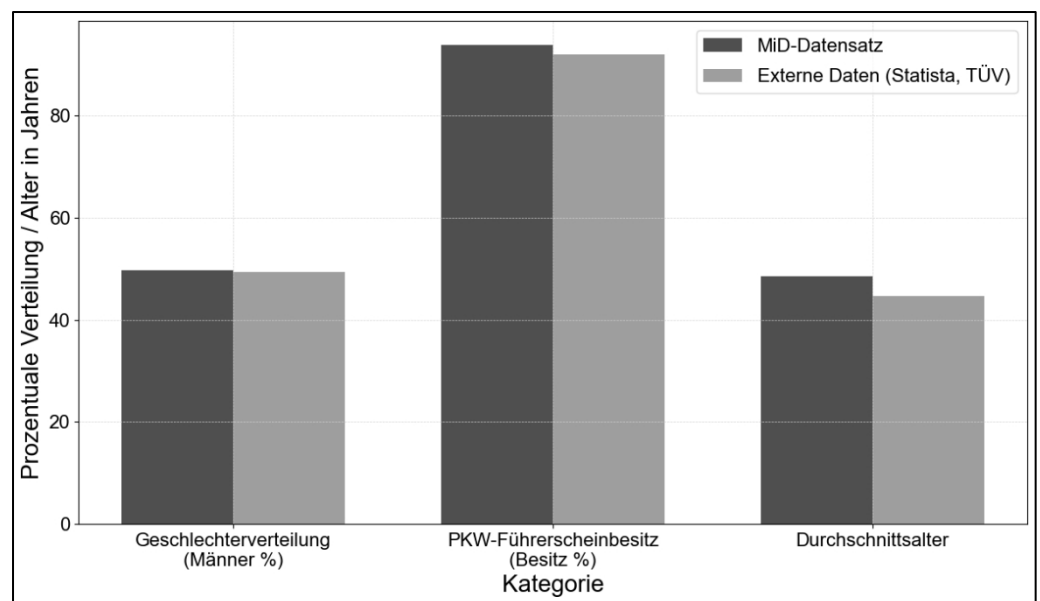


Figure 3: Comparison of MiD data with external data sources.

With regard to gender distribution, the ratio of male to female participants is calculated. The result shows a distribution of 49.69% men to 50.31% women. For comparison: According to Statista, the gender distribution is 49.30% male and 50.70% female (Statista, 2024a). Furthermore, the MiD dataset is analyzed with regard to the possession of car driving licenses, whereby the age group 0 to 17 years is excluded. This analysis shows that 93.82% of participants have a car driving license, while 6.18% do not. In comparison, the TÜV association reports that 92% of adult German citizens have a car driving license (Bühler, 2022). The average age in the MiD study was determined to be 48.6 years, while according to Statista it is 44.6 years (Statista, 2024b). This discrepancy could be due to various

factors. One possible reason is the specific composition of the sample. It has a higher number of participants in the older age groups, as these may be easier to reach in telephone interviews.

There is an overrepresentation of urban households in the survey, which is mainly due to the different types of contact (Nobis, 2019, p. 36). The data is adjusted to accurately reflect the age and gender distribution in each area, based on the official census data. This means that the information on the age and gender of the people corresponds exactly to the demographic data of the respective survey area as recorded in the census (Nobis, 2019, p. 50).

The plots illustrate the availability and use of bicycles and pedelecs. Germany is divided into 5 x 5 km grid cells⁴. To enable this representation, the 5 x 5 km grid is first converted into coordinates (see Appendix A.3). The visualized grid cells do not reflect participation in the survey everywhere. As a result, some areas on the map remain blank and appear as "white spots" (e.g. in parts of Brandenburg). In contrast, white grid cells with clear grid boundaries describe 0% bicycle availability.

⁴5 x 5 km cell division follows the INSPIRE division.

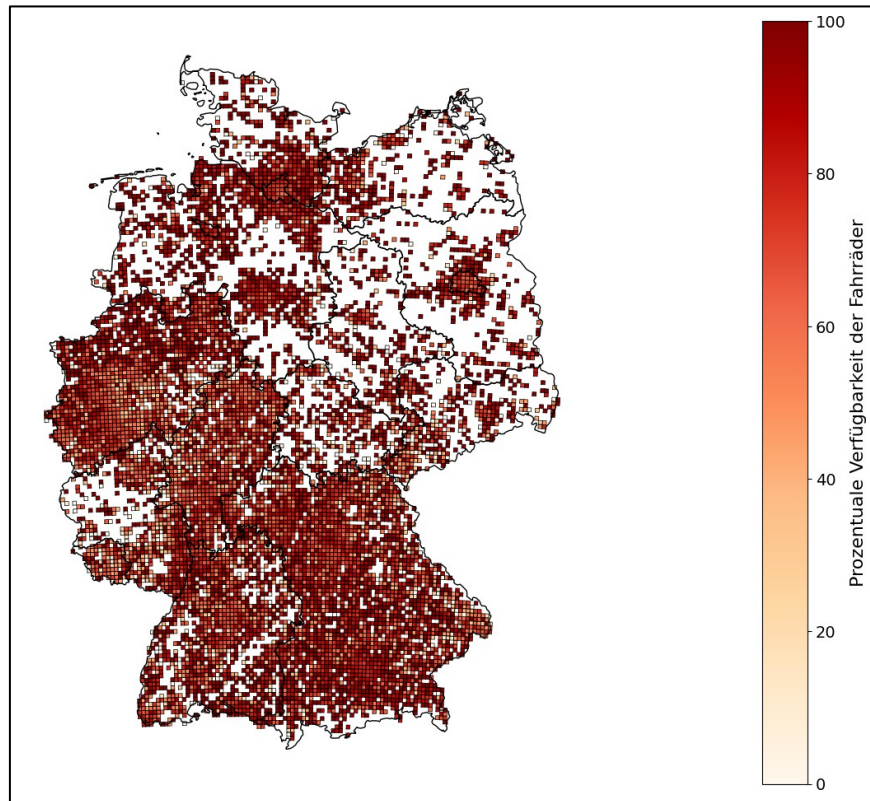


Figure 4: Availability of bicycles and pedelecs in Germany, based on MiD dataset.

Figure 5 shows the ratio (N_g) of the actual use of bicycles (N) to the total number of respondents (G). The ratio is calculated using sliding (9) is calculated. Specifically, the study examines how many people use their bicycle at least once a week.

$$N_{\#} = \frac{N}{G} \times 100. \quad (9)$$

A more differentiated picture emerges here, with a visible reduction in usage density compared to availability. This underlines the discrepancy between the possibility of using a bicycle and the actual decision to do so.

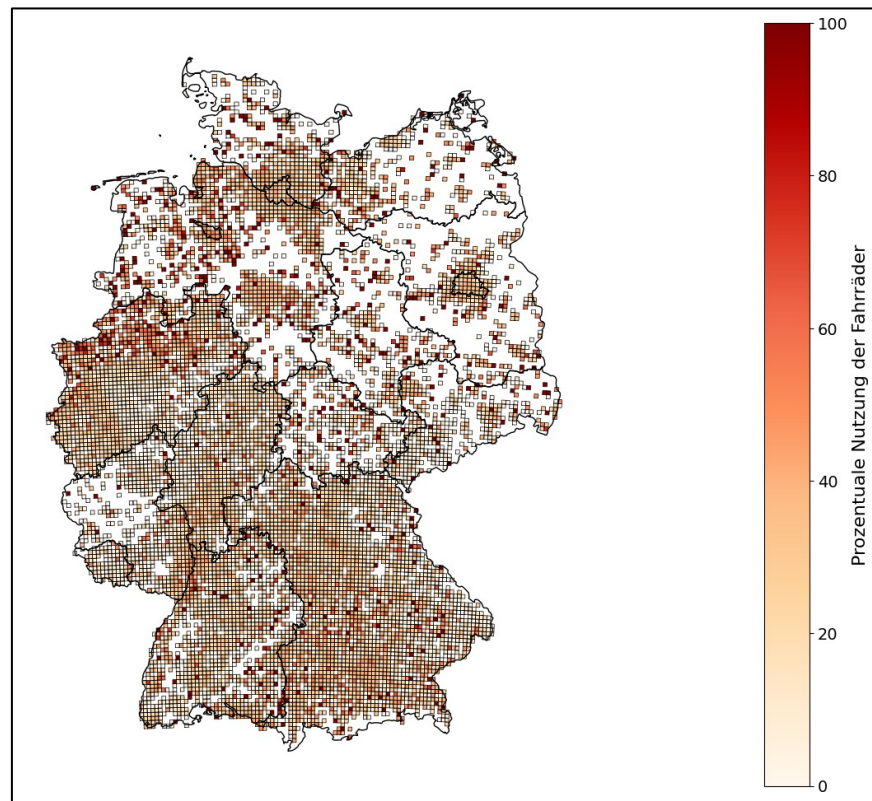


Figure 5: Actual use of bicycles and pedelecs in Germany, based on MiD data set.

5.3 Logistic regression model

A logistic regression model is used to analyze the influence of bicycle infrastructure on bicycle use behavior in urban and rural areas. Logistic regression makes it possible to model the probability of an event occurring and to explain how different independent variables (predictors) affect a binary dependent variable. This is useful for investigating cause-and-effect relationships (Stoltzfus, 2011, p. 1100).

In this context, this study analyzes the decision between using a bicycle or pedelec and a car. The data set is filtered so that both a bicycle or pedelec and a car are available for each trip. This analysis focuses on trips within grid cells in which the survey participation comprises at least 20 people and a minimum participation rate of 0.01% of the population per grid cell is achieved.

The 5 x 5 km grid cells are divided into rural and urban grid cells. Rural regions are characterized by a population density of less than

than 300 inhabitants per km² (Dörzenbach et al., 2017, p. 10). For the purpose of this analysis, it is assumed for the sake of simplicity that a population density of over 300 inhabitants per km² characterizes an urban region. Georeferenced population data for 2021, divided into 1 x 1 km grid cells, is used for the calculation. On this basis, the population is determined within 5 x 5 km grid cells. Which corresponds to 1,230 grid cells in rural areas and 1,433 grid cells in urban areas. The analysis focuses on distances between 0 and 10 km. This is due to the fact that 71 % of the distances traveled by bicycle and pedelec were less than 3 km, which roughly corresponds to the radius of a 5 x 5 km cell (see Figure 9 in Appendix A.5). Finally, in the logistic regression model

31,140 trips by bicycle, 2,566 trips by pedelec and 91,341 trips by car were analyzed.

Variable type	Variable	Possible values
Dependent variable	Means of transportation: bicycle/pedelec or car	0= Car 1= Bicycle/Pedelec
Independent variables (binary)	Season: Winter	0= no, 1= yes
	Precipitation	0= no, 1= yes
	Purpose: transportation or work	0= no, 1= yes
	Accompaniment	0= no, 1= yes
	Gender: Man	0= no, 1= yes
	High economic status	0= no, 1= yes
	High level of education	0= no, 1= yes
Independent variables (ordinal)	Age groups Distance	0 - 8 (0 =< 18, 8 => 80)
	Bikeability Index	0 - 10 [km] 0 - 10

Table 11: List of variables of the logistic regression model.

One-hot coding is used to convert the data extracted from the MiD data set into a binary system. The binary variables are: Season: winter, precipitation, route purpose, company, gender: male, high economic status and high level of education. In addition, the ordinal variables such as age, distances and the final rating of the Bikeability Index and its individual indicators are included in the study (see Table 11). In the analysis, it is crucial to categorize the various factors into dependent and independent variables. The

The dependent variable is the choice of means of transport, which indicates whether someone chooses the bicycle/pedelec (= 1) or the car (= 0).

5.4 Results of the logistic regression model

This chapter presents the results of the logistic regression analysis. The results include various dimensions of analysis, starting with the evaluation of the basic model (see Table 12), followed by a separate analysis of urban and rural areas (see Table 18 in Appendix A.5). Furthermore, the study includes an analysis of the indicators and their comparison between rural and urban regions (see Table 13). Finally, the analysis includes both a separate consideration of the pedelec (see Table 19 in Appendix A.5) and an examination of the influence of the bicycle infrastructure on the distance traveled (see Table 20 in Appendix A.5). The results, including the estimated coefficients, standard errors, z-values, p-values and 95% confidence intervals, are listed for each independent variable. The test accuracy of the basic model of 67 % indicates that it makes precise predictions in a majority of cases. The recall value for class 1 (0.69) implies that 69% of all bicycle trips are correctly identified by the model (see Table 16 in Appendix A.5). Furthermore, there is no multicollinearity between the independent variables, which rules out the possibility that independent variables are strongly correlated with each other and influence the result (see Table 15 in Appendix A.5).

	Coefficient	Std. Err.	z-value	p-value	[0,025	0,975]
Bikeability Index	0,316	0,004	86,017	< 0,001	0,309	0,323
Season: Winter	-0,843	0,018	-48,079	< 0,001	-0,877	-0,809
Precipitation	0,055	0,019	2,956	0,003	0,019	0,091
Purpose: Transportation/Work	-0,683	0,014	-50,113	< 0,001	-0,710	-0,656
Male	-0,111	0,013	-8,912	< 0,001	-0,136	-0,087
Age	-0,077	0,004	-21,892	< 0,001	-0,083	-0,070
High econom. Status	-0,175	0,013	-13,762	< 0,001	-0,200	-0,150
High level of education	0,281	0,013	21,115	< 0,001	0,255	0,307
Distance [km]	-0,270	0,003	-102,852	< 0,001	-0,275	-0,264

Table 12: Results of the logistic regression.

	Coefficient	Std. Err.	z-value	p-value	[0,025	0,975]
Bikeability Index (Total)	0,316	0,004	86,017	< 0,001	0,309	0,323
Bikeability Index (country)	0,298	0,004	72,187	< 0,001	0,290	0,306
Bikeability Index (Urban)	0,283	0,010	28,086	< 0,001	0,264	0,303
Main street ⁽⁵⁾ (Total)	0,017	0,003	5,69	< 0,001	0,011	0,023
Main road (country)	0,014	0,014	2,30	0,022	0,002	0,026
Main street (Urban)	0,018	0,004	4,97	< 0,001	0,011	0,024
Topography (total)	0,231	0,007	34,05	< 0,001	0,217	0,244
Topography (land)	0,340	0,014	24,23	< 0,001	0,312	0,367
Topography (Urban)	0,213	0,008	26,84	< 0,001	0,198	0,229
Surface (total)	-0,119	0,007	-16,66	< 0,001	-0,133	-0,105
Surface (land)	-0,066	0,014	-4,63	< 0,001	-0,093	-0,038
Surface (Urban)	-0,162	0,008	-19,54	< 0,001	-0,179	-0,146
Traffic-calmed ⁽⁶⁾ (Total)	0,034	0,006	5,32	< 0,001	0,021	0,046
Traffic-calmed (country)	0,079	0,014	5,70	< 0,001	0,052	0,106
Traffic-calmed (Urban)	0,063	0,007	8,83	< 0,001	0,049	0,078
Connectivity (total)	0,174	0,004	45,40	< 0,001	0,166	0,181
Connectivity (land)	0,009	0,027	0,33	0,740	-0,044	0,062
Connectivity (Urban)	0,171	0,004	39,05	< 0,001	0,162	0,179
Green space (total)	0,021	0,002	9,98	< 0,001	0,017	0,025
Green space (land)	0,019	0,005	3,97	< 0,001	-0,010	0,029
Green space (Urban)	0,025	0,002	10,77	< 0,001	0,020	0,029

Table 13: Results of the logistic regression to determine the influence of the individual factors of the Bikeability Index.

5.5 Discussion of the results

The results of the model in Table 12 show that the log odds of bicycle use increase by 0.316 units for every unit increase in the final score of the Bikeability Index. This illustrates the positive influence of bicycle infrastructure on bicycle use behavior. The statistical significance and precision of this observation is underpinned by the p-value (< 0.001) and the confidence interval (0.309 to 0.323). Figure 8 in the appendix

A.4 visualizes the Bikeability Index in Germany and shows that grid cells with a high rating also tend to have a high level of bicycle use.

⁵Proportion of cycle paths on main roads.

⁶Proportion of traffic-calmed and bicycle-friendly streets.

When looking at the other factors, it is noticeable that the level of education influences the choice of cycling as a means of transport with a comparable intensity (0.281). This observation suggests that measures to sensitize the population to the positive effects of bicycle use could play just as important a role as improving the bicycle infrastructure. In contrast, people with a higher economic status (-0.175) are less inclined to choose the bicycle. With increasing distance (-0.270), the probability of using a bicycle decreases, although this effect is less pronounced than the influence of cycling infrastructure.

The positive coefficient for precipitation (0.055) could generally be influenced by the low number of trips in the rain by bicycle ($n = 4518$). This figure may indicate a group of cyclists who cycle consistently regardless of the weather and could therefore influence the statistical analysis (see Figure 10 in Appendix A.5). In contrast, however, it is clear that the seasonal conditions in winter (-0.843) have the greatest influence on bicycle use.

While the study by Codina et al. (2022, p. 7) found a positive correlation between men and cycling, the present model shows that women (-0.111) have a stronger tendency to use bicycles. This could indicate that men prefer the car more often for the trips analyzed (see Table 17 in Appendix A.5). The model shows that age (-0.077) has a slight influence on the general use of bicycles and pedelecs. When considered separately, the use of pedelecs increases significantly with age (coefficient= 0.293; $p < 0.001$), while that of bicycles decreases (coefficient= -0.113; $p < 0.001$). This indicates that older people tended to prefer pedelecs in 2017, a trend that may have changed since then. For more extensive transportation tasks and transporting passengers, the car is usually preferred. These circumstances are not taken into account in the current model.

In rural (0.283) and urban (0.298) areas, the Bikeability Index shows a positive influence of bicycle use (see Table 18 in Appendix A.5). The influence is slightly higher in urban areas than in rural areas. This indicates that infrastructural improvements in cities potentially have a stronger influence on increasing bicycle use

than in more rural areas. This could be due to the fact that distance has a greater negative influence in rural areas (-0.295). The reason for this is the distances traveled in rural areas, which are longer on average (3.92 km compared to 3.64 km in urban areas).

The analysis of the indicators in Table 13 shows that surface quality (-0.119) has a negative influence on cycling behavior. However, when surface condition is considered in isolation, a slight positive influence is revealed (coefficient = 0.004; $p < 0.001$). Overall, this indicates that the surface quality has no influence on the choice of means of transport according to this model. On the one hand, this is in line with the research by Mertens et al. (2016, p. 11), who identify other indicators as more important. On the other hand, Arellana et al. (2020, p. 326) classify this as the strongest indicator. One possible reason for this could be that Arellana et al. determine the surface condition based on the detection of potholes and unevenness. An analysis that is worth recommending in future work.

The proportion of cycle paths along main roads shows a consistent but weak influence on cycling behavior in both rural (0.014) and urban (0.017) regions. Furthermore, the proportion of traffic-calmed and bicycle-friendly roads (0.034) and the presence of green spaces (0.021) exert a minor influence, which proves to be similar in rural and urban areas. In contrast, Codina et al. (2022, p. 4) and Hardinghaus et al. (2021, p. 11) were able to show that these indicators have an influence in large cities. However, the subdivision of rural and urban regions based on a population density of 300 inhabitants per km² does not allow for this analysis.

It can be seen that connectivity in rural areas (0.009) only has a marginal influence. However, the width of the confidence interval (-0.044 to 0.062) together with the p-value (0.740) indicates a low statistical significance of this estimate. In contrast, the influence in urban areas (0.171) is clear and statistically significant. Codina et al. (2022, p. 8) confirm that, regardless of the quality and safety of cycle paths, a poorly connected network ultimately makes cycling less practical.

Topography has a stronger influence on bicycle use in rural regions (0.340) than in urban regions (0.213). The analysis of the data shows a significant difference between the average ratings of topography in urban (7.136) compared to rural (6.297) areas. Consequently, rural areas have a greater gradient on average. In the model studied, topography (0.231) is the dominant factor in the decision to choose cycling as a mode of transport (see Table 13). This result is in line with previous research (Codina et al., 2022, p. 8; Krenn et al., 2015, p. 455), while other studies cite topography as an important factor but not the most important one (Arellana et al., 2020, p. 325; Winters et al., 2013, p. 878).

The importance of topography underlines the challenge posed by hilly terrain. Pedelects are a viable alternative in such areas. A comparison of the factors between bicycles and pedelecs reveals that a lower gradient score (-0.243) increases the propensity to choose pedelecs over bicycles (see Table 19 in Appendix A.5). In addition, distance (0.235) has a lower influence on the use of pedelecs, indicating that pedelec users tend to travel longer distances than bicycle users. The analysis shows that both the surface quality (0.068) and the proportion of traffic-calmed and bicycle-friendly roads (0.089) are more significant indicators for pedelec use than for bicycle use. In contrast, connectivity (-0.053) and the proportion of green and water areas (-0.032) have a stronger influence on bicycle use compared to pedelec use.

The investigation of the correlation between the Bikeability Index and the distance covered in the range of 5 to 10 kilometers shows that a higher rating of the bicycle infrastructure (0.063) correlates positively with the willingness to ride longer distances by bicycle or pedelec (see Table 20). However, this positive association appears marginal compared to the sharp decline in bicycle use in winter (-0.573), which highlights the significant decrease in willingness to ride such distances in colder months.

Previous studies (Arellana et al., 2020, p. 323; Hardinghaus et al., 2021, p. 10; Krenn et al., 2015, p. 456; Winters et al., 2013, p. 874) have already

The results show that the concept of the Bikeability Index includes several dimensions and consists of a large number of indicators. The results emphasize how crucial the multidimensional approach is, as only the aggregated index has significant predictive power, while individual indicators do not. In summary, it can be stated that the cycling infrastructure has a significant influence on cycling behavior in urban and rural regions. However, cycling infrastructure is only one of several important factors.

5.6 Limitations of the model

The limitations of the model are discussed below. Firstly, no information is available on the selected routes within the 5 x 5 km grid cell, which means that all segments in the grid cell are evaluated and not just the segments of the route traveled. Due to the evaluation over 5 x 5 km grid cells, the averaging of the results can lead to incorrect statements if there are large differences in the bicycle infrastructure within the cell. Secondly, there are limitations to the data sources. In particular, the challenge of uniformly labeling bicycle infrastructure with OSM, as well as the overlapping of green and water areas, leads to limitations. In addition, the MiD data from 2017 may no longer reflect the current situation. Thirdly, an important trade-off in this research concerns the balance between level of detail and transferability or practical applicability. The focus is on transferability and applicability, which means that not all factors are taken into account.

6 Conclusion

This research project dealt with the question of what influence bicycle infrastructures have on the usage behavior of bicycles and pedelecs in rural and urban regions of Germany. The methodological approach included the identification and weighting of relevant bicycle infrastructure indicators. The following indicators were identified: Proportion of traffic-calmed roads, connectivity of cycle paths, proportion of cycle paths on main roads, proportion of green and water areas, topography and surface conditions. The indicators were integrated into a Bikeability Index using OSM data and its influence was measured by means of a logistic regression analysis using MiD data. The research results show that the bicycle infrastructure has a significant influence on the decision to use a bicycle in rural and urban areas. The gradient has the greatest influence on the Bikeability Index for bicycles, but not for pedelecs. Furthermore, the results show that bicycle and pedelec users travel longer distances with a better bicycle infrastructure.

The use of OSM as a data source proved to be particularly valuable for this work. Its great advantage lies in the timeliness and accessibility of a wide range of data on cycling infrastructure. The mobility study highlighted the different factors influencing the use of bicycles and pedelecs. However, the study made it necessary to use 5 x 5 km grid cells for the analysis. This prevented the analysis of specific cycle routes.

The findings of this work offer concrete points of reference for urban planners and political decision-makers to guide investments in bicycle infrastructure in a targeted manner and thus promote bicycle use. The ability to evaluate cycling infrastructure automatically and always up to date opens up prospects for comprehensive analyses of the links between cycling infrastructures and aspects such as public health or accident rates. Future research could also take a closer look at pedelecs in order to investigate their potential for expanding bicycle mobility.

Bibliography

- Arellana, J., Saltarín, M., Larrañaga, A. M., González, V. I. & Henao, C. A. (2020). Developing an urban bikeability index for different types of cyclists as a tool to prioritize bicycle infrastructure investments. *Transportation Research Part A: Policy and Practice*, 139, 310-334. <https://doi.org/10.1016/j.tra.2020.07.010>
- Bocksch, R. (June 3, 2020). The best cycling cities in Germany. *Statista*. <https://de.statista.com/infografik/21890/die-besten-fahrradstaedte-germany/>
- Boeing, G. (2017). OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65, 126-139. <https://doi.org/10.1016/j.compenvurbsys.2017.05.004>
- Boeing, G. (2023). OSMnx-Documentation. *Computers, Environment and Urban Systems*, 1-141.
- Boucher, M. (2019). Transportation Electrification and Managing Traffic Congestion: The role of intelligent transportation systems. *IEEE Electrification Magazine*, 7(3), 16-22. <https://doi.org/10.1109/mele.2019.2925730>
- Brown, B. B., Smith, K. R., Hanson, H., Fan, J. X., Kowaleski-Jones, L. & Zick, C. D. (2013). Neighborhood design for walking and biking: physical activity and body mass index. *American journal of preventive medicine*, 44(3), 231-238. <https://doi.org/10.1016/j.amepre.2012.10.024>
- Bühler, J. (2022). *Driving license as important as ever - test gets top marks from drivers*. VdTÜV - Verband der TÜV e.V. <https://www.tuev-verband.de/pressemitteilungen/fuehrerscheinumfrage> [Accessed on: 10.01.2024]
- Chan, E. Y. C. & Cooper, C. H. V. (2019). Using road class as a replacement for predicted motorized traffic flow in spatial network models of cycling. *Scientific reports*, 9(1), 19724. <https://doi.org/10.1038/s41598-019-55669-8>
- Codina, O., Maciejewska, M., Nadal, J. & Marquet, O. (2022). Built environment bikeability as a predictor of cycling frequency: Lessons from Barcelona. *Transportation Research Interdisciplinary Perspectives*, 16. <https://doi.org/10.1016/j.trip.2022.100725>
- Dörzenbach, A., Link, T. & Schneider, E. (2017). The German Urban Audit - Comparison of Cities in the European Statistical System. *KOSIS-Community Urban Audit*.
- Eggs, Johannes. Follmer, Robert. Gruschwitz, Dana. Nobis, Claudia. Bäumer, Marcus. Pfeiffer & Manfred (2019). Mobility in Germany - MiD method report. BMVI, infas, DLR, IVT, infas 360. Bonn, Berlin. *Study conducted by infas, DLR, IVT and infas 360 on behalf of the*

Federal Ministry of Transport and Digital Infrastructure. Bonn, Berlin.
https://www.mobilitaet-in-deutschland.de/archive/pdf/MiD2017_Analyse_zum_Rad_und_Fussverkehr.pdf

- Ferster, C., Fischer, J., Manaugh, K., Nelson, T. & Winters, M. (2020). Using OpenStreetMap to inventory bicycle infrastructure: A comparison with open data from cities. *International Journal of Sustainable Transportation*, 14(1), 64-73.
<https://doi.org/10.1080/15568318.2018.1519746>
- Fishman, E. & Cherry, C. (2016). E-bikes in the Mainstream: Reviewing a Decade of Research. *Transport Reviews*, 36(1), 72-91.
<https://doi.org/10.1080/01441647.2015.1069907>
- Gallo, M. & Marinelli, M. (2020). Sustainable Mobility: A Review of Possible Actions and Policies. *Sustainability*, 12(18), 7499.
<https://doi.org/10.3390/su12187499>
- Gullón, P., Badland, H. M., Alfayate, S., Bilal, U., Escobar, F., Cebrecos, A., Diez, J. & Franco, M. (2015). Assessing Walking and Cycling Environments in the Streets of Madrid: Comparing On-Field and Virtual Audits. *Journal of Urban Health*, 92(5), 923-939.
<https://doi.org/10.1007/s11524-015-9982-z>
- Hardinghaus, M. & Nieland, S. (2021). Assessing cyclists' routing preferences by analyzing extensive user setting data from a bike-routing engine. *European Transport Research Review*, 13(1), 1-19.
<https://doi.org/10.1186/s12544-021-00499-x>
- Hardinghaus, M., Nieland, S., Lehne, M. & Weschke, J. (2021). More than Bike Lanes-A Multifactorial Index of Urban Bikeability. *Sustainability*, 13(21), 11584. <https://doi.org/10.3390/su132111584>
- Hartanto, K., Anna Beatriz, G., M.F.A.M. van Maarseveen & M.J.G. Brussel (2017). Developing a bikeability index in the context of transit-oriented development (TOD). In *15th International Conference on Computers in Urban Planning and Urban Management (CUPUM)*.
https://www.researchgate.net/profile/anna-beatriz-grigolon/publication/318792598_developing_a_bikeability_index_in_the_context_of_transit-oriented_development_tod
- Heinen, E., Maat, K. & van Wee, B. (2013). The effect of work-related factors on the bicycle commute mode choice in the Netherlands. *Transportation*, 40(1), 23-43. <https://doi.org/10.1007/s11116-012-9399-4>
- Huber, S., Lisner, S., Lindemann, P., Muthmann, K., Schnabel, A. & Friedl, J. (2021). Modeling bicycle route choice in German cities using open data, MNL and the bikeSim web-app. In *2021 7th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)*. IEEE. <https://doi.org/10.1109/mt-its49943.2021.9529273>

- Ito, K. & Biljecki, F. (2021). Assessing bikeability with street view imagery and computer vision. *Transportation Research Part C: Emerging Technologies*, 132, 103371. <https://doi.org/10.1016/j.trc.2021.103371>
- Krenn, P. J., Oja, P. & Titze, S. (2015). Development of a Bikeability Index to Assess the Bicycle-Friendliness of Urban Environments. *Open Journal of Civil Engineering*, 05(04), 451-459. <https://doi.org/10.4236/ojce.2015.54045>
- Litman, T. (2013). *Evaluating active transport benefits and costs. Guide to valuing walking and cycling improvements and encouragement programs*. <https://scholar.google.de/citations?user=o-mo3saaaaaj&hl=en&oi=sra>
- Marshall, W. E. & Garrick, N. W. (2011). Research Article: Evidence on Why Bike-Friendly Cities Are Safer for All Road Users. *Environmental Practice*, 13(1), 16-27. <https://doi.org/10.1017/S1466046610000566>
- Mertens, L., van Dyck, D., Ghekiere, A., Bourdeaudhuij, I. de, Deforche, B., van de Weghe, N. & van Cauwenberg, J. (2016). Which environmental factors most strongly influence a street's appeal for bicycle transport among adults? A conjoint study using manipulated photographs. *International Journal of Health Geographics*, 15(1), 31. <https://doi.org/10.1186/s12942-016-0058-4>
- Nankervis, M. (1999). The effect of weather and climate on bicycle commuting. *Transportation Research Part A: Policy and Practice*, 33(6), 417-431. [https://doi.org/10.1016/s0965-8564\(98\)00022-6](https://doi.org/10.1016/s0965-8564(98)00022-6)
- Nobis, C. (2019). Mobility in Germany - MiD. Analyses of cycling and walking. *Study by infas, DLR, IVT and infas 360 on behalf of the Federal Ministry of Transport and Digital Infrastructure. Bonn, Berlin*. https://www.mobilitaet-in-deutschland.de/archive/pdf/MiD2017_Analyse_zum_Rad_und_Fussverkehr.pdf
- OpenStreetMap. (2022). *OpenStreetMap Deutschland - Die freie Wiki-Weltkarte*. <https://www.openstreetmap.de/> [Accessed on: 05.12.2023]
- OpenStreetMap Wiki: Map Features. (2022). *Map features - OpenStreetMap Wiki*. https://wiki.openstreetmap.org/wiki/Map_features [Accessed on: 12.02.2024]
- OpenStreetMap: ODbL. (2022). *Open Database License - OpenStreetMap Wiki*. https://wiki.openstreetmap.org/wiki/Open_Database_License [Accessed on: 05.12.2023]
- Pucher, J. & Buehler, R. (2008). Making Cycling Irresistible: Lessons from The Netherlands, Denmark and Germany. *Transport Reviews*, 28(4), 495-528. <https://doi.org/10.1080/01441640701806612>
- Sallis, J. F., Cerin, E., Conway, T. L., Adams, M. A., Frank, L. D., Pratt, M., Salvo, D., Schipperijn, J., Smith, G., Cain, K. L., Davey, R., Kerr, J., Lai, P.-C., Mitáš, J., Reis, R., Sarmiento, O. L., Schofield, G., Troelsen, J., van Dyck, D., . . . Owen, N. (2016). Physical activity in

- relation to urban environments in 14 cities worldwide: a cross-sectional study. *Lancet (London, England)*, 387(10034), 2207-2217. [https://doi.org/10.1016/S0140-6736\(15\)01284-2](https://doi.org/10.1016/S0140-6736(15)01284-2)
- Schröder, A. (2021). Fahrradstraßen 2.0: More space and attention for cycling in Münster. *Standort*, 45(2), 77-82. <https://doi.org/10.1007/s00548-021-00709-7>
- Sears, J., Flynn, B. S., Aultman-Hall, L. & Dana, G. S. (2012). To Bike or Not to Bike. *Transportation Research Record: Journal of the Transportation Research Board*, 2314(1), 105-111. <https://doi.org/10.3141/2314-14>
- Statista. (2024a). *Population shares by gender and generation 2022* | Statista. <https://de.statista.com/statistik/daten/studie/1131309/umfrage/bevoelkerungsanteile-nach-geschlecht-und-generation/> [Accessed on: 10.01.2024]
- Statista. (2024b). *Average age of the population in Germany by 2022* | Statista. <https://de.statista.com/statistik/daten/studie/1084430/umfrage/durchschnittsalter-der-bevoelkerung-in-deutschland/> [Accessed on: 10.01.2024]
- Steinacker, C., Storch, D.-M., Timme, M. & Schröder, M. (2022). Demand-driven design of bicycle infrastructure networks for improved urban bikeability. *Nature Computational Science*, 2(10), 655-664. <https://doi.org/10.1038/s43588-022-00318-w>
- Stinson, M. A. & Bhat, C. R. (2003). Commuter Bicyclist Route Choice: Analysis Using a Stated Preference Survey. *Transportation Research Record*, 1828(1), 107-115. <https://doi.org/10.3141/1828-13>
- Stoltzfus, J. C. (2011). Logistic regression: a brief primer. *Academic Emergency Medicine*, 18(10), 1099-1104. <https://doi.org/10.1111/j.1553-2712.2011.01185.x>
- Teixeira, I. P., Da Rodrigues Silva, A. N., Schwanen, T., Manzato, G. G., Dörrzapf, L., Zeile, P., Dekoninck, L. & Botteldooren, D. (2020). Does cycling infrastructure reduce stress biomarkers in commuting cyclists? A comparison of five European cities. *Journal of Transport Geography*, 88, 102830. <https://doi.org/10.1016/j.jtrangeo.2020.102830>
- Valenzuela, A. L. E. M., Lopes, A. A. d. S., Araujo, P. A. B. de, Della Justina, M. D., Arins, G. C. B. & Rech, C. R. (2022). Geospatial indicators of bikeability index as cyclefriendly city design: a systematic review. *Revista Brasileira de Atividade Física & Saúde*, 27, 1-12. <https://doi.org/10.12820/rbafs.27e0255>
- Walk Score. (2024). *Bike Score Methodology*. www.walkscore.com/bike-score-methodology.shtml <https://doi.org/10.1000/1.1000> [Accessed on: 18.02.2024]
- Willis, D. P., Manaugh, K. & El-Geneidy, A. (2015). Cycling Under Influence: Summarizing the Influence of Perceptions, Attitudes, Habits, and Social Environments on Cycling for Transportation. *International Journal of*

Sustainable Transportation, 9(8), 565-579.
<https://doi.org/10.1080/15568318.2013.827285>

Winters, M., Brauer, M., Setton, E. M. & Teschke, K. (2013). Mapping bikeability: a spatial tool to support sustainable travel. *Environment and Planning B: Planning and Design*, 40(5), 865-883.
<https://doi.org/10.1068/b38185>

Wysling, L. & Purves, R. S. (2022). Where to improve cycling infrastructure? Assessing bicycle suitability and bikeability with open data in the city of Paris. *Transportation Research Interdisciplinary Perspectives*, 15, 100648. <https://doi.org/10.1016/j.trip.2022.100648>

Appendix

A.1 Overview of the state of science

Author	Year	Indicators	Methods	Data sources	Place	Limitation
Codina et al.	2022	Traffic, road separation, connectivity, bicycle parking, topography	1) Local survey (n=290) to obtain indicators 2) Logistic regression to test the correlation between bicycle friendliness and choice of bicycle	Local survey, official Barcelona traffic survey, GIS data	Barcelona	Barcelona only
Hardinghaus et al.	2021	Spread of neighborhood streets, street networking, bicycle facilities along main roads, green paths, other facilities	1) Literature analysis of the indicators 2) Expert survey for weighting (n=57) 3) OpenStreetMap implementation 4) Multinomial logit model analyzes influence of bicycle friendliness on travel behavior	Expert survey, OSM, Berlin household survey	Berlin	Substrate is not evaluated (although considered important by experts)
Winters et al.	2013	Length of cycle paths, cycle paths separated from motorized traffic, connectivity of cycle-friendly roads, topography, density of destinations	1) Literature analysis of the indicators 2) Opinion survey (n=1402), traffic analyses (n=3280) and expert surveys (n=23) to determine weighting 3) Spatial tool to visualize bikeability	Vancouver bicycle survey, GIS data	Vancouver	Spatial Tool does not offer up-to-dateness
Krenn et al.	2015	Length of cycle paths, separated cycle paths, main roads without parallel cycle paths, green and water areas, topography	1) cycle routes (n=278) to obtain indicators. 2) Validity of the index is measured with correlation of cycling behavior in Graz. 3) Mapping of bicycle friendliness	Mobility data from study, GIS	Graz	Mobility data not up to date
Hartanto et al.	2017	Traffic conditions, connectivity, infrastructure, environment, topography, bicycle parking spaces	1) Literature analysis of the indicators 2) Investigation of transit-oriented infrastructure at 21 stations	ArcGIS, Spatial Data from DutchCyclingUnion	Arnhem	Simplification of the weighting

Ito and Biljecki	2021	34 indicators divided into connectivity, environment, infrastructure, perception, vehicle-cyclist interaction	1) Literature analysis of the indicators 2) Use of StreetView(SV) and Computer Vision (CV) for evaluation	StreetView, OSM, air quality indices, ComputerVision, digital elevation models (DEM), Land use data	Singapore and Tokyo	Variances between the different data sources, SV data not always accurate as it is from the car's point of view. Simplification of the weights of the supercategories
Gullón et al.	2015	Surface, safety, aesthetics, cycle paths, cycle lanes	1) Interviews and Delphi study to determine indicators 2) Virtual and physical investigation of infrastructure segments (n=500).	Audit tools (M-SPACES), StreetView, On-Field Audit Tools, Survey	Madrid	No consideration of green spaces, Street View not so up-to-date
Arellana et al.	2020	Directness, comfort, coherence, attractiveness, safety	1) Literature analysis of the indicators 2) Discrete choice model with preferences, perceptions and socio-economic characteristics to develop index 3) Weighting estimated from survey (n=336)	Surveys among cyclists, ArcGIS	Barranquilla (Colombia)	-
Wysling and Purves	2021	Speed limit, cycle path infrastructure, topography	1) Literature analysis of indicators for identifying opportunities to improve cycling infrastructure 2) Calculation of bikeability based on open data	OSM, urban infrastructure data (Paris)	Paris	No consideration of connectivity, underground quality, traffic volume
Stinson and Bhat	2003	Riding time, surface quality, cycle paths, separated cycle paths, parking spaces	1) Surveys on commuting by having to choose between 2 routes (n=3145) 2) Analysis of the survey to determine route decision preferences recognize	Surveys	-	Only computer survey, only survey to bicycle organization

Table 14:

Research status of the Bikeability Index.

A.2 Road networks from OSMnx

The OSM data model consists of nodes (points), edges (connecting nodes) and relations that define relationships between collections of nodes and edges. To assign specific properties or meanings to these elements, OSM uses a system of tags (Ferster et al., 2020, p. 3). A tag is a key-value pair ("*key*" = "*value*") that is assigned to an element to describe it in more detail, such as the type of road ("*highway*"= "*residential*").

To simplify the collection, creation and analysis of road networks using OSM data, the Python library OSMnx is used. This application offers five key functions: automated download of political boundaries and building contours, user-defined creation of road network data from OSM, correction of network topology, storage of road networks in various formats and comprehensive network analysis. This includes route calculation, visualization and the evaluation of metric and topological properties of the network (Boeing, 2017, p. 126).



Figure 6: Various road networks of the OSMnx output.

Figure 6 is the result of mapping generated using OSMnx. The section shown was chosen specifically to show the diversity of the

urban routes in a defined space. The illustration is based on a resolution of 500 x 500 meters.

It is possible to select different network types to determine which road segments should be covered. In particular, the representation of *"all_private"* paths in blue illustrates the additional paths that go beyond public thoroughfares and could potentially be relevant for emergency services, private driveways or special access points (Boeing, 2017, p. 131). In contrast, the *'all'* network, highlighted in green, provides an overview of paths intended for general use, without the restrictions of private paths (Boeing, 2017, p. 131). Finally, the orange *"bike"* network shows the permeability for cyclists. In the research presented here, the focus is on examining the bicycle infrastructure exclusively in areas that are permitted for cyclists (Boeing, 2017, p. 131). Thus, routes tagged as *"motorway"*, *"motorway_link"*, *"trunk"* and *"trunk_link"* are not considered (Steinacker et al., 2022, p. 661). Therefore, the network setting remains fixed to the *"bike"* option in order to ensure a targeted and relevant analysis.

A.3 Conversion of the grids into coordinates

The conversion from a 5 x 5 km grid to a coordinate is carried out using a Python function. The transformation is carried out from the Lambert projection (EPSG:3035) to the usual geodetic reference system (ESPG:4326), thereby obtaining the south-western coordinate of the grid. Next, a 5 x 5 km grid is developed from the converted coordinate. Since one degree of latitude on the earth's surface corresponds approximately to 111320 meters, the change in latitude is calculated using equation (10).

$$\Delta lat = \frac{5000}{111320}. \quad (10)$$

The calculation of the change in longitude in equation (11) is more complex, since the length of a longitude varies with the cosine of the latitude.

$$\Delta lon = \frac{5000}{* 40075000 \cos(rad(latitude)) / 360}. \quad (11)$$

Equation (12) is used to determine the geographical coordinates of all the boundaries of a grid and the final development of the grid:

$$\begin{aligned} &= north\ latitude + \Delta lat. \\ &= south\ latitude. \\ &west = longitude + \Delta lon. \\ &east = longitude. \end{aligned} \quad (12)$$

A.4 Visualization of the Bikeability Index

Figure 7 shows the use of bicycles in relation to the number of people surveyed. Usage values are filtered out where the percentage of respondents was less than 0.1 % per grid cell and at least 20 people were surveyed. This filters out unrepresentative values due to low participation. This representation therefore provides a more accurate picture of bicycle use. It is particularly evident that Münster, known as a cycling city (Schröder, 2021, p. 78), is characterized by a high percentage of bicycle use.

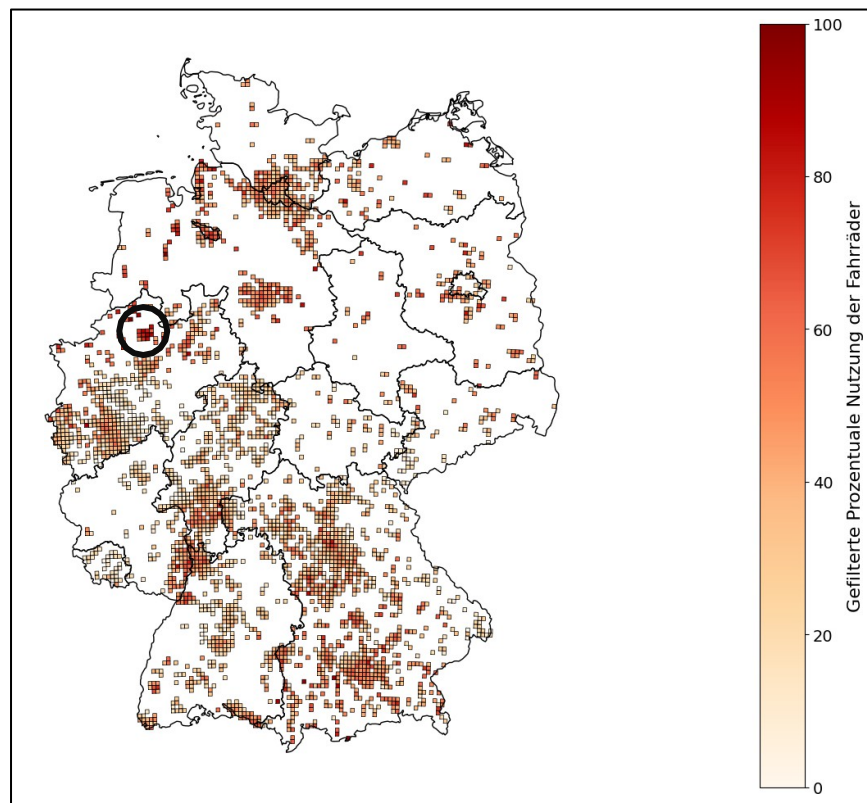


Figure 7: Filtered use of bicycles and pedelecs with marking from the city of Münster.

Figure 8 visualizes the quality of cycling infrastructure in various German grid cells, with a higher score indicating better infrastructure quality.

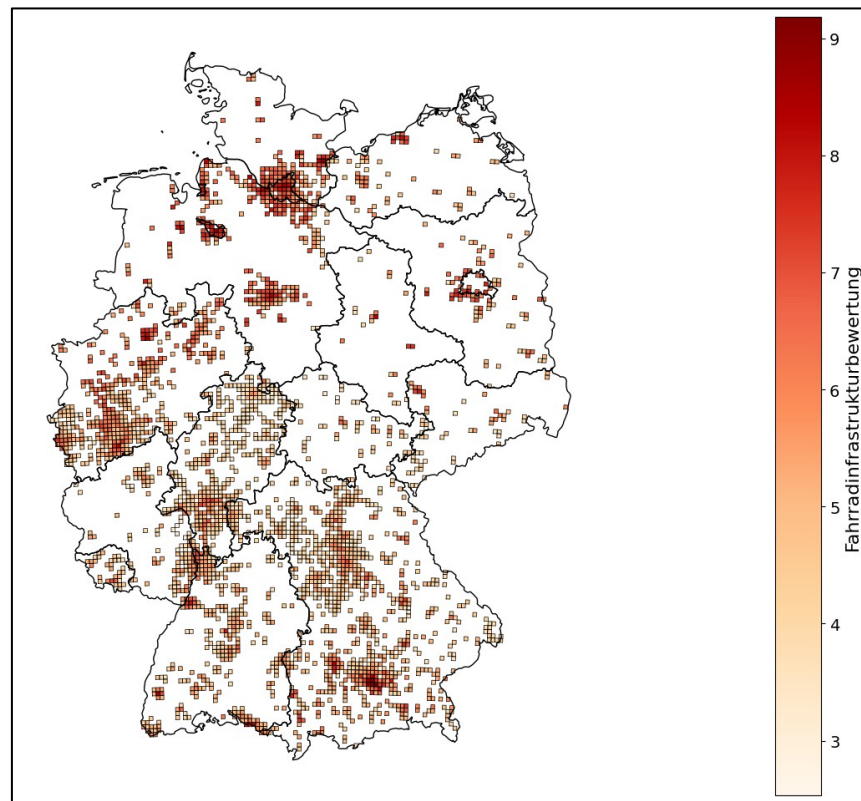


Figure 8: Bikeability index of the filtered grid cells.

A.5 Model results

	Valuance Inflation Factor
Constant	43,115
Bikeability Index	1,031
Season: Winter	1,009
Precipitation	1,009
Purpose: Transportation	1,006
Male	1,048
Age	1,144
High econom. Status	1,139
High level of education	1,147
Distance	1,010

Table 15: Multicollinearity of the factors.

	Precision	Recall	F1 score
Class 0	0,67	0,65	0,66
Class 1	0,66	0,69	0,68
Accuracy (Accuracy)			0,67

Table 16: Precision, Accuracy, Recall of the model.

Paths [%]	Men	Women
Car	75,97 %	72,99 %
Bicycle/Pedelec	24,03 %	27,01 %

Table 17: Percentage distribution of routes based on gender.

Urban vs. rural	Coefficient	Std. Err.	z-value	p-value	[0,025	0,975]
Bikeability Index (Urban)	0,298	0,004	72,187	< 0,001	0,290	0,306
Bikeability Index (country)	0,283	0,010	28,086	< 0,001	0,264	0,303
Season: Winter (Urban)	-0,779	0,019	-40,210	< 0,001	-0,817	-0,741
Season: Winter (Country)	-1,363	0,045	-30,357	< 0,001	-1,451	-1,275
Precipitation (Urban)	0,076	0,021	3,666	< 0,001	0,035	0,116
Precipitation (land)	0,043	0,042	1,033	0,302	-0,039	0,126
Transportation/Labor (Urban)	-0,640	0,015	-41,482	< 0,001	-0,670	-0,610
Transportation/Labor (Country)	-1,027	0,029	-35,166	< 0,001	-1,085	-0,970
Male (Urban)	-0,118	0,014	-8,355	< 0,001	-0,145	-0,090
Male (country)	-0,015	0,027	-0,543	0,587	-0,068	0,039
Age (Urban)	-0,084	0,004	-21,200	< 0,001	-0,092	-0,077
Age (country)	0,100	0,008	11,944	< 0,001	0,084	0,117
High econom. Status (Urban)	-0,171	0,014	-11,848	< 0,001	-0,199	-0,143
High econom. Status (country)	-0,189	0,027	-6,970	< 0,001	-0,242	-0,136
High level of education (Urban)	0,328	0,015	21,560	< 0,001	0,298	0,358
High level of education (country)	0,094	0,028	3,393	< 0,001	0,039	0,148
Distance (Urban)	-0,270	0,003	-89,384	< 0,001	-0,276	-0,264
Distance (country)	-0,295	0,005	-54,834	< 0,001	-0,306	-0,285

Table 18: Results of the logistic regression in urban and rural regions.

Pedelec vs. bicycle	Coefficient	Std. Err.	z-value	p-value	[0,025	0,975]
Bikeability Index	-0,125	0,029	-4,290	< 0,001	-0,183	-0,068
Main street ⁷	0,001	0,020	0,052	0,959	-0,039	0,041
Topography	-0,243	0,049	-5,002	0,000	-0,338	-0,148
Surface	0,068	0,050	1,373	0,170	-0,029	0,166
Traffic-calmed ⁸	0,089	0,046	1,934	0,053	-0,001	0,180
Connectivity	-0,053	0,027	-1,943	0,052	-0,107	0,001
Green space	-0,032	0,015	-2,041	0,041	-0,062	-0,001
Season: Winter	-0,673	0,148	-4,553	0,000	-0,962	-0,383
Precipitation	0,218	0,140	1,561	0,118	-0,056	0,491
Purpose: Transportation/Work	0,099	0,096	1,024	0,306	-0,090	0,288
Male	-0,130	0,094	-1,378	0,168	-0,314	0,055
Age	0,053	0,038	1,392	0,164	-0,022	0,129

⁷Proportion of cycle paths on main roads.⁸Proportion of traffic-calmed and bicycle-friendly streets.

High econom. Status	0,085	0,102	0,830	0,407	-0,116	0,286
High level of education	0,236	0,100	2,348	0,019	0,039	0,433
Distance	0,235	0,022	10,906	0,000	0,193	0,278

Table 19: Results of the logistic regression of the pedelec vs. bicycle comparison.

Distance (5-10km)	Coefficient	Std. Err.	z-value	p-value	[0,025	0,975]
Bikeability Index	0,063	0,005	11,715	< 0,001	0,052	0,073
Season: Winter	-0,573	0,036	-16,171	< 0,001	-0,643	-0,504
Precipitation	-0,049	0,031	-1,612	0,107	-0,109	0,011
Purpose: Transportation/Work	-0,597	0,021	-28,453	< 0,001	-0,638	-0,556
Male	0,464	0,021	22,581	< 0,001	0,423	0,504
Age	-0,055	0,005	-10,017	< 0,001	-0,062	-0,044
High econom. Status	0,104	0,021	4,927	< 0,001	0,063	0,145
High level of education	0,025	0,023	1,091	0,275	-0,020	0,070

Table 20: Results of the logistic regression for the use of bicycles and pedelecs over a distance of 5 - 10 km.

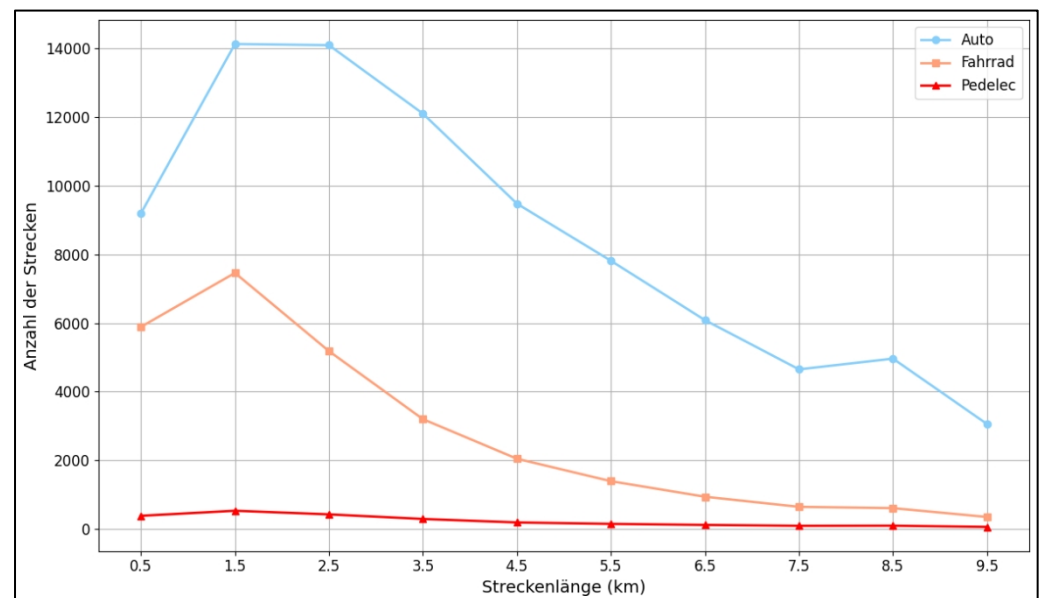


Figure 9: Comparison of the frequency of use of cars, bicycles and pedelecs over different route lengths.

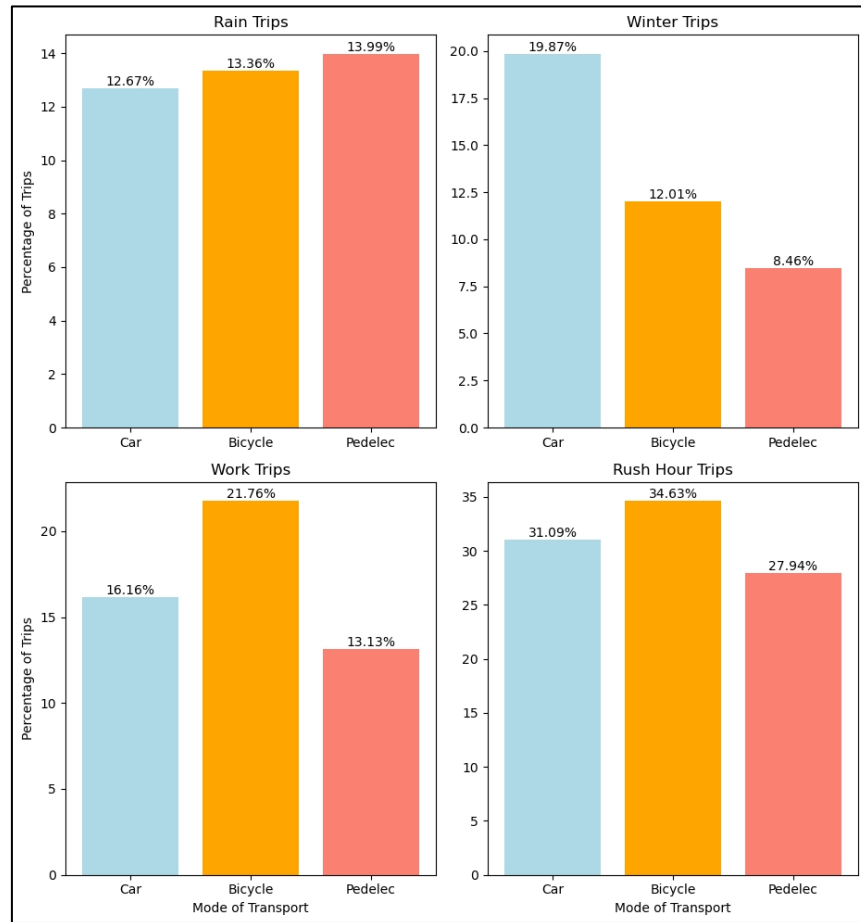


Figure 10: Usage rates of cars, bicycles and pedelecs under different conditions: Rain, winter, work and rush hour.

Declaration of honor

I hereby declare that I have written this thesis without unauthorized assistance and without the use of aids other than those specified. I have marked all passages taken verbatim or in spirit from publications as such. The submitted Bachelor's thesis has not been the subject of another examination procedure, neither in its entirety nor in essential parts. The electronic version of the submitted Bachelor's thesis is identical in content and formatting to the paper copy.

Freiburg, March 18, 2024

First and last name